Tactical Clinical Text Mining for Improved Patient Characterization

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***Abstract*—Clinical sources of information are markedly increasing in both volume and variety. A significant portion of the valuable data resides in the unstructured or semi-structured clinical text of documents stored in disparate repositories or embedded in HL7 messages. Clinical documents such as discharge summaries, prescriptions, lab reports, and free-form physician notes are filled with abbreviations, acronyms, misspellings, and ungrammatical phrases. However, synoptic reporting methods are restrictive for health care practitioners who wish to express critical and comprehensive patient information in electronic medical records. Furthermore, they have been superseded by systems that use natural language processing (NLP) to extract clinical concepts from free-form text. To address the growing need for efficient NLP solutions that can handle the volume and variety of clinical text, we have developed an optimized rules-based clinical concept extractor called TRACE (Tactical Rules-based AQL Clinical Extractor) using the Annotation Query Language (AQL). We present the experience we have gained applying text mining tools to this challenging domain, as well as a comparison of our solution to cTAKES (clinical Text Analysis and Knowledge Extraction System), an open-source clinical text miner, on a set of prescription documents. We also describe how efficient and scalable clinical text mining techniques will improve several of our company's offerings.**

Keywords-Text Mining, Natural Language Processing, Clinical Data, UIMA, cTAKES, SystemT, Annotation Query Language

I. INTRODUCTION

The majority of clinical records consists of unstructured and semi-structured clinical text and contains exclusive medical and social insights into a patient's health. Stored in disparate clinical repositories and exchanged via Health Level 7 (HL7) messages, this information is embedded in a variety of sources such as patient history records, lab reports, prescriptions, pathologies and discharge summaries. For example, Fig. 1 shows an excerpt from an HL7 message containing a patient prescription. While some fields contain standard codes, many fields contain variable abbreviations and unstructured text (see Section VII for further discussion of the variability of HL7 messages).

Not only is clinical information multifaceted, but it is massive. Specifically, the widespread adoption of the electronic medical record (EMR) and the clinical document architecture has greatly increased the volume of clinical narrative. For instance, in 2011 various clinical document repositories in Alberta contained 300 million documents, and this total was estimated to grow by over 55 million documents per year [1]. Therefore, extracting clinical concepts from unstructured text is a 'big data' challenge that calls for a resourceful and efficient solution. Conventional technologies that use the Common Pattern Specification Language (CPSL) or a pipeline of analysis engines are ill-equipped for massive-scale text mining tasks. For instance, cTAKES (clinical Text Analysis and Knowledge Extraction System) is a comprehensive open source Natural Language Processing (NLP) pipeline system designed to extract and map information from EMRs to clinical concept codes [2]. cTAKES can take 24 hours to process a mere 90,000 clinical notes with an average size of 80 lines using entry-level hardware (a single core 2GHZ processor and 2 GB of physical memory) [3]. Therefore, we present a novel solution for clinical concept extraction built on SystemT, a technology that explicitly optimizes the execution strategy of rules written in the Annotation Query Language (AQL) and that is integrated with IBM's BigInsights, which is built on top of Hadoop [4]. SystemT has been used in financial [5] and social media applications [6], and for patient smoking status classification [7]. However, our clinical extractor TRACE (Tactical Rules-based AQL Clinical Extractor) finds clinical concepts codes, and has modules for patient demographics, lab results, and prescriptions.

```
AL1|||99999998^No Known Drug Allergies
DG1||||||A
ORC|XO|0000010|||IP||1^BID&1000,2200,^^200910150932^^0^0^
RXE|1^BID&1000,2200,^^200910150932^^0^0^|361906^PROPRANOLOL 40MG TAB(INDERAL)|40||MG|EACH|HOLD FOR SBP #lg;90 |||1||||||||||||
RXR|^PO
OBX|1|ST|1010.3^Height||072|Inches
OBX|2|ST|1010.1^Body Weight||190.00|pounds
```

Figure 1. Excerpt from an HL7 message used to place an order with a pharmacy or dispensing system [8]. Allergy (AL1), diagnosis (DG1), common order (ORC), pharmacy encoded order (RXE), route (RXR), and observation (OBX) segments may contain structured, semi-structured, or unstructured data.

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First, in Section II we describe the underlying algorithms of traditional rules-based annotators, pipeline approaches, and SystemT. In Section III and IV we introduce cTAKES and TRACE, respectively. We compare cTAKES and TRACE on a set of prescription documents in Section V, and discuss our plans for future improvements to TRACE in Section V.D. Section VI includes general lessons that we have learned from applying NLP tools to clinical text. Finally, in Section VII we discuss our plans to integrate TRACE with other innovative data science projects.

II. Text Mining Background

Many traditional rules-based NLP approaches use CPSL. With CPSL, text is annotated in a stream from left to right, and rules are cascaded together from the bottom-up [9]. However, this leads to significant performance and flexibility issues. On the other hand, SystemT is designed for multifaceted extraction tasks that are efficient on very large data sets [4]. SystemT uses a declarative language called AQL to translate queries into algebraic expressions that operate on document text, and facilitates a hierarchy of annotations to be created from the results.

SystemT overcomes four major problems with CPSL systems: lossy sequencing, rigid matching, and limited expressivity and performance [9]. With CPSL systems, rules are evaluated as regular expressions over a sequence of input symbols. Therefore, when spans of overlapping text qualify for the *same* rule, arbitrary tie-breakers must be imposed to supply a non-overlapping input stream for the next higher-level phase of rule matching. However, with AQL, the spans of input or output annotations are allowed to overlap or a specific consolidation policy may be employed to ensure the desired outcome. Furthermore, matches for *different* rules may also overlap, whereas with CPSL a particular span of text may only match one rule, which is a gross limitation. AQL also offers significant improvements in the ability to express complex rule interactions and perform more sophisticated extraction tasks over CPSL. Finally, SystemT uses extensive rule rewrites and cost-based optimization to improve efficiency. With rules-based approaches, run time is most significantly influenced by operations like regular expression and dictionary matching [9]. CPSL systems use an engine that supports the full POSIX regular expression standard for every rule, and examine every character in the input document stream for regular expression matches. They also impose a rigid rule evaluation order. By contrast, SystemT uses specialized regular expression engines wherever possible, shares work across different rules (especially for dictionary matching), and optimizes the order of rule evaluations, allowing some evaluations to be skipped entirely.

An alternative to a hierarchical annotation scheme is a pipeline of analysis engines, where the output annotations of an upstream module become the input annotations for the next extraction task. In this way, a train of specialized modules including both rules-based and machine-learning components may be used to process a document. For instance, cTAKES [2] is built upon the Unstructured Information Management Architecture (UIMA) and implements both rules-based modules and machine learning components from OpenNLP [10–12]. The machine learning components have been trained on manually annotated clinical datasets consisting of unstructured text from EMRs. Therefore, each is designed to improve the accuracy of the resulting annotations compared to a manually constructed set of rules. However, a major limitation of the pipeline approach is that it constitutes a static, non-optimized execution plan. All annotators in the pipeline are run consecutively. By contrast, SystemT employs a pragmatic, relationship-centric approach. For instance, SystemT will skip a rule evaluation entirely if the possibility of a higher-level annotation that joins two annotation types is precluded by the fact that the first rule evaluation produced no document annotations [4]. Although cTAKES lacks this tactical pre-emptive optimization, it is notable that some have attempted to improve efficiency by deploying it on a cloud-based computing platform [3].

III. Overview of cTAKES

The cTAKES clinical text mining system includes a full pipeline of annotators that have been tested at both the component and system levels [2]. We discuss a few significant modules here.

A. Normalization

cTAKES uses the Lexical Variant Generation tool of the Unified Medical Language System (UMLS) to normalize each word token with respect to inflections, conjugations, word order in multi-word terms, alphabetic case, punctuation, and possessives [13]. This improves the recall of named entities since both the normalized and non-normalized forms are used for dictionary look-up.

B. Part-of-speech Tagging and Constituency Parsing

cTAKES uses a maximum entropy-based part-of-speech (POS) tagger and a constituency parser from the OpenNLP toolkit [10]. Together these form a syntactic parse tree for each sentence that represents the likely semantic interpretation. Each derivation of a parse tree is scored using a maximum-entropy model [12]. The probability, $p_X(a|b)$, of each action a in a procedure, $X \in$ {TAG,CHUNK,BUILD, CHECK}, for a partial derivation context b is learned from the training set. The TAG procedure assigns POS tags in a first pass over the input sentence. In the second pass, CHUNK creates chunks from POS tags and other words in order to create "flat" phrases such as a noun phrase. The third pass alternates between BUILD and CHECK, forming higher-level phrases and checking if further joins are possible. Then, the score of the parse tree T with derivation, $\{a_1, \ldots, a_n\}$, is assigned as

$$\text{score}(T) = \prod_{a_i \in \text{deriv(T)}} q(a_i|b_i), \quad (1)$$

where

$$q(a_i|b_i) = \begin{cases} p_{\text{TAG}}(a|b) & \text{if } a \text{ is an action from TAG} \\ p_{\text{CHUNK}}(a|b) & \text{if } a \text{ is an action from CHUNK} \\ p_{\text{BUILD}}(a|b) & \text{if } a \text{ is an action from BUILD} \\ p_{\text{CHECK}}(a|b) & \text{if } a \text{ is an action from CHECK} \end{cases} . \quad (2)$$

The search for the best parse tree is performed by maximizing the score using a breadth-first search.

C. Named Entity Recognition

cTAKES uses a dictionary that is a subset of the UMLS Metathesaurus for named entity recognition (NER) [14]. The noun phrases identified by the constituency parser provide the lookup window for the dictionary matching algorithm [2]. The head and modifiers within the noun phrases are permutated and checked against the dictionary. The NER component also maps named entities to their corresponding UMLS concept codes. The semantic types of the dictionary terms can also be configured. Typical classes include disorders and diseases, signs and symptoms, procedures, anatomy, and drugs.

D. Negation

When extracting information from clinical texts, it is not only important to identify medical concepts, but also to ascertain their asserted polarity. cTAKES uses the NegEx pattern-matching algorithm to detect negation near named entities [15]. With NegEx, negation phrases are first identified by manually examining a training set. The algorithm is then used to search for these negation phrases near named entities in a test set. Sentences that produced no match in the test set are then isolated and scanned for other negation phrases that were missed. An advantage of NegEx over its predecessors is that it limits the scope of the negation phrase and thus more accurately identifies the target. Also, it filters out "pseudo-negation" phrases such as double-negatives that are misleading.

E. Drug Profile Module

cTAKES includes a special pipeline that extracts drug references and attributes from clinical narratives [16]. The upstream NER module first extracts drug references using a special configuration of the UMLS dictionary, and returns RxNorm concept unique identifiers (CUIs) for various types such as the Ingredient (IN), Semantic Clinical Drug (SCD), and Semantic Branded Drug (SBD) types [17]. Note that this module does not include a list of abbreviations used in daily practice. Rules for extracting drug attributes are applied to windows defined by new line characters and sentence boundaries, which are useful for list-type formats and narratives, respectively. If multiple drugs are mentioned in the same window with a basic sequential drug reference and attribute pattern, the attributes are assigned to the closest reference. In more complicated situations, the same attributes may be assigned to more than one drug reference. The drug attribute annotations include *Strength*, *Form*, *Route*, *Dose*, *Frequency*, *Duration*, *Start date*, *End date*, and *Change status*. *Change status* indicates if a medication prescription has changed (increased, decreased, started or stopped).

IV. Overview of TRACE

TRACE is a clinical text extractor that implements tactical rule rewrites, workload sharing, and rule order optimization using SystemT. TRACE has modules for patient demographic information, lab reports, and prescriptions. In this paper, we present the prescription extractor module.

A. Dictionary

TRACE uses the RxTerms database [18] for drug name lookup. RxTerms was created as an aid for writing prescriptions or medication histories and is a subset of the RxNorm vocabulary provided free of cost by the U.S. National Institutes of Health. An advantage of using RxTerms is that it includes separate fields for drug name, synonyms and abbreviations, strength, form, and route. Drug ingredients are also provided, which have no associated attributes. We removed terms from our dictionary that had long composite drug names or were retired. We used the first word of the drug name to look for matches in candidate text and defined these as anchor words. In future work, we will define a lookup window around the anchor word and create permutations of this word with other tokens from the window to match the full name from the dictionary, similar to the approach used by cTAKES (see Section V.D).

B. Base-level Extraction Tasks

TRACE produces a set of concept annotations using a hierarchy of rules. The base-level rules include dictionary and regular expression matching. We defined several dictionaries containing terms and abbreviations commonly used in clinical texts. For drug names, we used a dictionary created from the RxTerms database (see above). However, to prevent generic matches, we explicitly removed certain words from the output annotations that were either contained in an exclusion list imported from cTAKES or were manually added. For the drug form dictionary, we used distinct values from the RxTerms database in addition to some common abbreviations. Other base-level annotations consist of verbs, numbers (digits, words, and adverbs), time references (units, adverbs), strength and dose units, route terms, frequency terms, and duration terms. Also, some rules were created using regular expressions to match code-like frequency abbreviations such as "Q8S".

C. Drug Attribute Annotations

Annotations are abstracted above the base-level with AQL rules that implement pattern matching. For example, a *Strength* annotation is a number reference immediately followed by a strength unit. In many cases, we also included a regular expression to look for concatenations of numbers

and units with no space in between. Other drug attribute annotations include *Form*, *Route*, *Dose*, *Frequency*, *Duration, Time of day*, *and Instruction*. *Instruction* refers to an administration instruction such as "take" or "swallow". To handle cases where exact or non-exact overlapping spans of text match more than one rule for the same attribute, the 'ContainedWithin' consolidation policy is used to filter out inferior spans of text. If overlapping spans of text match rules for different attributes, each respective attribute retains the matching text. This is a highly effective, unbiased approach.

D. Intermediate Annotations

A *Medication mention* annotation is created by associating a *Strength* and *Form* annotation with a drug reference. *Strength* and *Form* are particularly important since they are used to look up RxNorm CUIs. These attributes commonly occur close to the drug name, and are thus expected to follow the drug name or each other within two tokens.

A *Prescription* annotation consists of a common pattern of drug attributes. Either a *Frequency* or a *Route* annotation is required, while *Instruction*, *Dose*, *Time of day* and *Duration* are optional. For example, the extraction pattern shown in Fig. 2 matches the prescription "2 TAB Every 8 hours X 2 Weeks(s)", as well as "BID PO". *Duration* is allowed to follow the other annotations within two tokens, since we have observed that it occasionally occurs non-adjacent to the other attributes. Overlapping spans of text that match a certain prescription pattern, as well as those that qualify for more than one prescription rule are consolidated using the 'ContainedWithin' consolidation policy.

E. Head Annotations

TRACE creates top-level annotations that join *Medication mention* annotations with prescription patterns and RxTerms information. A *Medication prescription* annotation is defined where a prescription pattern follows a *Medication mention* within seven tokens. This distance is selected based on observations from our data, but in the future we will impose more robust boundaries (see Section V.D). A *Medication info* annotation is created by performing an inner join of tuples from the *Medication mention* structure with a table created from the RxTerms database where the drug name (the lookup word) and the *Strength* and *Form* attributes are linked (see Fig. 3). This provides a valid RxNorm CUI for each identified IN, SCD, or SBD reference.

V. Comparison of cTAKES and TRACE

In this section, we provide a comparison of the cTAKES drug profile module to the TRACE prescription extractor module. Both cTAKES and TRACE identify drug references from the RxNorm vocabulary [17] and common prescription annotations including *Strength*, *Form*, *Route*, *Dose*, *Frequency*, and *Duration*.

```
--Frequency required
create view Prescription1 as
extract pattern (<I.match>)?(<D.match>)?(<F.match>)
      (<R.match>)?(<T.match>)?<Token>{0,2}(<L.match>)?
return group 0 as match
      and group 1 as instruction
      and group 2 as dose
      and group 3 as frequency
      and group 4 as route
      and group 5 as timeOfDay
      and group 6 as duration
from Instruction I, Dose D, Frequency F, Route R,
      TimeOfDay T, Duration L
consolidate on match using 'ContainedWithin';
```

Figure 2. Example of a pattern extraction AQL statement from TRACE. A question mark indicates that the annotation is optional.

```
--To identify the CUI, both strength and form must match
create view MedicationInfo as
select M.match, M.FirstWord, M.Strength, M.Form, T.RXCUI
from MedicationMention M, RxTerms T
where Equals(M.FirstWord, T.LOOKUP_WORD)
      and Equals(M.Strength, T.STRENGTH)
      and Equals(M.Form, T.NEW_DOSE_FORM);
```

Figure 3. AQL statement showing an inner join of TRACE annotations with RxTerms data to identify the RxNorm CUI.

A. Data

For this comparison, we prepared several HL7 messages, a sample physician note distributed with cTAKES, and a de-identified prescription document (see Table I). The HL7 messages are order messages (RDE-O01 or RDE-011) that are typically dispatched to a pharmacy and/or dispensing system. We chose fields from each message that could be used to represent the essential elements of the prescription, and concatenated them to form a string. Taken at face-value, this operation creates unstructured data from structured fields, but in reality the fields contain a mixture of drug types (ingredients, brands, and generic references), semi-codified abbreviations, and other unnormalized synonyms, abbreviations, and free-form administration instructions. The physician note was obtained from the sample data that was included with the cTAKES-2.5 release. The prescription PDF document was converted to plain text using the Nuance SDK for Optical Character Recognition (OCR) [19].

B. Methods

1) Gold Standard Annotations

To compare the drug references and attributes extracted by cTAKES and TRACE, we created a gold standard annotation set. For this task, we only considered drug types and attributes common to both cTAKES and TRACE. For drug types, this includes the IN, SCD and SBD types. To match an SCD or SBD type, both a *Strength* and *Form* attribute is required. To match an IN type, neither *Strength* nor *Form* may be included. Note that cTAKES outputs both an SCD or SBD type and an IN type when a *Strength* and *Form* annotation are associated with a drug reference, since overlapping annotations are allowed. Similarly, TRACE would output one *Medication mention* annotation that

TABLE I. PRESCRIPTIONS DOCUMENTS

Document [source]	Text
1 [HL7]	IBUPROFEN 100 MG/5ML PO SUSP. ONCE X1 PO.
2 [HL7]	PROPRANOLOL 40MG TAB(INDERAL). BID PO. HOLD FOR SBP #lg;90
3 [HL7]	NACL 0.9%. Q8S PB. TOTAL VOLUME = 60 ML INFUSE OVER 30 MINUTES
4 [HL7]	FENTANYL INJ. ONCE IV. Once ROUTINE ; X 30 Days.
5 [HL7]	alendronic acid 70 mg tablet, 4. D28 ORA37. Take 1 tablet by mouth every Saturday . Swallow whole. Take with a full glass of water at least 30 mins before food or 2 hours after food. Remain upright for at least 30 mins after taking tablet.
6 [cTAKES sample note]	IMPRESSION/REPORT/PLAN The patient has very typical history for claudication. Yet the patient does not have any significant bruit at the pelvic area, abdominal area. Aspirin 200-mg once-a-day. Ibuprofen increased from 200 mg to 300 mg p.o. twice a day. Aspirin 80 mg. acetaminophen discontinued.
7 [OCR-processed document]	Date Printed 05/10/2011 REPRINTED Rx cortisone 25 mg tablet DIN 84037940 Take 0.5 TAB Once daily for 30 days Qty: 45 tablet Route: oral Refills: 1 Srart: 05/10/2011 11/06/2011 NEW Rx TYLENOL EXTRA STRENGTH 500 MG TABLET DIN 09838908 2 TAB Every 8 hours for 2 Weeks Qty: 84 tablet Route: oral Refills: 2 srart: 05/10/2011 06/21/2011

includes *Strength* and *Form* and one that does not. Therefore, our gold standard annotation set contains overlapping *Medication mention* annotations. However, if only a *Strength* attribute was linked to a drug reference (with no *Form* attribute), it was omitted from the gold standard since this would represent a Semantic Clinical Drug Component (SCDC) or a Semantic Branded Drug Component (SBDC) type, which are not included in the RxTerms database (but are supported by cTAKES). Also, when more than one distinct span of text qualifying for a drug attribute could be related to a certain drug reference, all possible combinations of the drug reference with the matches were annotated.

2) Metrics

Several commonly employed metrics for text mining include precision (positive predictive value), recall (sensitivity), F-score (the harmonic mean of precision and recall), and accuracy, defined as:

$$precision = \frac{TruePositives}{TruePositives + FalsePositives}, \quad (3)$$

$$recall = \frac{TruePositives}{TruePositives + FalseNegatives}, \quad (4)$$

$$F\text{-}score = \frac{2 \cdot (precision \cdot recall)}{(precision + recall)}, \quad (5)$$

$$accuracy = \frac{TruePositives + TrueNegatives}{TruePositives + FalsePositives + FalseNegatives + TrueNegatives}. \quad (6)$$

For drug references (*Medication mention* annotations), we evaluated the F-score, which is a good metric for balancing precision and recall. Also, when dealing with binary classification problems where the class (in this case a named entity) may be assigned to a span of text longer than a single token, there are numerous combinations of tokens that qualify for 'true negatives', and so it is considerably more difficult to evaluate the accuracy of an NER extractor on a long document. Furthermore, the class distribution is very skewed when only one or two named entities occur per document, and so the F-score is a more appropriate measure [20].

On the other hand, accuracy is a good metric for evaluating drug attribute annotations. Attributes are assigned based on clues from various text spans, but they are not output independently of drug references. In addition, cTAKES normalizes attribute annotations and does not output the qualifying text spans, thus the attributes may be considered labels. A true positive occurs if a drug reference with a certain attribute value is labelled correctly, and a true negative occurs if a drug reference with an empty attribute value in the gold standard is labelled with 'null' for that attribute. A false positive occurs if an incorrect label is assigned to a drug attribute, while a false negative occurs if no value is assigned to an attribute with a non-null value in the gold standard.

C. Results

Here we present results from the cTAKES drug profile module (version 3.1) and the TRACE prescription extractor for the document collection of Table I. The results are helpful for an investigation of the strengths and weaknesses of each extractor, but are not intended to be used for a rigorous evaluation of performance on large data sets.

Table II shows precision, recall, and F-score for *Medication mention* annotations. TRACE has a higher recall than cTAKES, resulting in a higher F-score. This is at least partially due to the fact that cTAKES outputs a set of normalized attribute labels per drug reference. If more than one span of text may be used to assign a certain attribute value, the gold standard includes all possible combinations of a drug reference with the text spans that qualify for the attribute. However, cTAKES does not output a separate *Medication mention* annotation unless there are clues that indicate different labels for the same attribute. Alternatively,

TRACE is configured to output each unique clue directly. Therefore, TRACE produces two out of the four possible gold standard "IBUPROFEN" ingredient annotations for Document 1, whereas cTAKES outputs only one. Similarly, TRACE finds three out of four possible "FENTANYL" annotations for Document 4, whereas cTAKES outputs only one. Still, this is not a critical failure of cTAKES, provided that the labels are accurate.

Table III shows the accuracy of the drug attribute annotations. TRACE shows a higher accuracy for the *Frequency* attribute compared to cTAKES. TRACE contains numerous terms and abbreviations in its frequency dictionary, and also searches for common word patterns and regular expressions that match composite frequency references like "Q8S". Similarly, TRACE uses regular expressions to detect some *Strength* annotations. For instance, TRACE annotates "100 MG/5ML", whereas cTAKES annotates the shorter span "100 MG" (see Table IV). cTAKES also misses the abbreviation "SUSP" for the *Form* attribute of this drug reference. Therefore, the annotation maps to an SCDC type instead of an SCD. Nevertheless, if we had considered the SCDC type in this study, annotating "100MG" instead of "100 MG/5ML" would have resulted in a false positive for both *Strength* and RxNorm CUI. On the other hand, TRACE is able to correctly identify the CUI.

TRACE also shows a higher accuracy for the *Dose* attribute. A weakness of the cTAKES drug profile module is that the rule for identifying *Dose* attributes is too ambiguous. The rule produces a match for any number reference that is close to a *Strength* or *Frequency* annotation. For instance, Table V shows that cTAKES assigns a Dose attribute of "45" to the "cortisone" drug reference instead of "0.5" or "0.5 TAB". Alternatively, TRACE creates a *Dose* annotation by searching for a number reference immediately followed by a dose unit. Thus, it correctly annotates "0.5 TAB" as the *Dose* attribute. However, the disadvantage of this approach is that for certain phrases where the number and dose unit are not adjacent, the *Dose* attribute would be missed. For example, TRACE would miss the *Dose* "two tablets" in the phrase "Take two 325-mg aspirin tablets".

Finally, we note that cTAKES is adept at assigning attributes when multiple drug references occur close together. For instance, cTAKES assigned "40MG" and "tablet" for the *Strength* and *Form* attributes, respectively, of the two drug references "PROPRANOLOL" and "INDERAL" from Document 2. In this case, the attributes are enclosed by two drug references in the same window. Therefore, cTAKES assigned the attributes to both drug references. On the other hand, TRACE does not yet consider patterns where drug attributes occur before drug references, or use multiple drug references to constrain the assignment of drug attributes.

TABLE II. NER RESULTS FOR *MEDICATION MENTION* ANNOTATIONS

	cTAKES	TRACE
Precision	1.0	1.0
Recall	0.35	0.74
F-score	0.52	0.85

TABLE III. ACCURACY OF DRUG ATTRIBUTE ANNOTATIONS

	cTAKES	TRACE
Strength	1.0	1.0
Form	1.0	1.0
Route	0.88	0.65
Dose	0.88	1.0
Frequency	0.57	0.82
Duration	0.88	0.88
RxNorm CUI	1.0	1.0

TABLE IV. SELECT ANNOTATIONS FROM DOCUMENT 1

	Gold Standard [span]	cTAKES [span/label]	TRACE [span]
Medication mention	IBUPROFEN 100 MG/5ML PO SUSP [0–28]	IBUPROFEN 100 MG [0–16]	IBUPROFEN 100 MG/5ML PO SUSP [0–28]
Strength	100 MG/5ML [10–20]	100MG [label]	100 MG/5ML [10–20]
Form	SUSP [24–28]	-	SUSP [24–28]
Route	PO [38–40]	Enteral Oral [label]	PO [38–40]
Dose	-	-	-
Frequency	X1 [35–37]	-	X1 [35–37]
Duration	-	-	-
RxNorm CUI (type)	197803 (SCD)	316072 (SCDC)	197803 (SCD)

TABLE V. SELECT ANNOTATIONS FROM DOCUMENT 7

	Gold Standard [span]	cTAKES [span/label]	TRACE [span]
Medication mention	cortisone [562–571]	cortisone [562–571]	cortisone [562–571]
Strength	-	-	-
Form	-	-	-
Route	oral [659–662]	-	-
Dose	0.5 TAB [606–613]	45 [label]	0.5 TAB [606–613]
Frequency	Once daily [614–624]	1day [label]	Once daily [614–624]
Duration	for 30 days [625–636]	for 30 days [label]	for 30 days [625–636]
RxNorm CUI (type)	2878 (IN)	2878 (IN)	2878 (IN)

D. Discussion

In this study, TRACE showed a higher NER F-score compared to cTAKES (see Section C). However, unlike cTAKES, TRACE is configured to output all possible combinations of drug references with attribute clues. Still, in the future, we will include normalization for TRACE's drug attribute annotations and also retain references to the text spans that triggered the labels. Also, note that although

cTAKES had a lower F-score than TRACE, it included annotations of more diverse semantic types from the UMLS dictionary (which were not included in the gold standard). In the future, we may broaden TRACE's dictionary to include these types, and also use any *Route* annotations associated with drug references to look up generic or branded doseform group concept codes. Additionally, we will implement a lookup algorithm that permutes candidate words against the drug dictionary since we expect that TRACE would perform poorly on long drug names without the ability to match word order variants. Furthermore, we should include rules for negation that discern when a drug is prescribed or recommended versus prohibited or discussed.

Our results also indicate that TRACE excels in identifying composite *Strength* and *Frequency* annotations compared to cTAKES. In addition, TRACE employs a more specific rule for identifying *Dose* annotations, resulting in improved accuracy. Still, in the future, we will slightly relax this rule by allowing some distance between the number reference and the dose unit. Furthermore, we have observed that cTAKES employs superior rules for attribute assignment when multiple drug references occur close together. Therefore, we will simplify TRACE's pattern matching to look for attributes within a certain distance of a drug reference, and use multiple drug references to impose boundaries. Thus, we will eliminate several rules consisting of permutations of long prescription patterns such as the one shown in Fig. 2. This should improve the recall of drug attributes. However, it would require that all intermediate-level annotations (in this case drug references linked with a particular drug attribute field) be 'outer' joined together to create a head annotation. While there is no explicit outer join operator in AQL, there is a syntactically complicated work-around.

VI. Lessons Learned in Clinical Text Mining

The benefits of clinical text mining are numerous. However, there are also several challenges that make data mining tasks particularly difficult in this domain. First, clinical text often contains short, ungrammatical phrases such as the ones we have presented in Table I. Although many of these documents were created by concatenating structured HL7 fields, they are fair representations of the typical short, code-like phrases often used by physicians and healthcare providers. For documents like these, we have found that a lightweight rules-based extractor produces high quality NER and attribute assignment results compared to a full-blown NLP pipeline. Clinical text also contains numerous abbreviations and acronyms that are often overloaded or misspelled. Furthermore, diverse sources of information including dictations, speech recognition applications, and OCR-processed documents introduce an additional source of error. Thus, we expect that a fuzzy matching algorithm would provide a more significant improvement in recall compared to a detailed semantic analysis such as is often performed by an NLP pipeline.

Clinical documents also often contain templates and pseudo-tables that make excessive use of whitespace [21]. We have found it relatively simple to mine this sort of data using the pattern extraction capabilities of AQL. For instance, TRACE includes a module for lab reports, which are often presented in tabular format using whitespace and new line characters. A hierarchical approach is an effective way to extract results since each row of the data follows a predictable pattern consisting of lower-level annotations such as test name, result, reference range, and units.

VII. Future Goals

In Section V.D we presented our plans to improve and extend several of TRACE's rules and annotations. Here we describe our plans to use TRACE to enhance our company's offerings. First, HL7 messages carry valuable clinical information between provincial/state clinical repositories, hospital information systems, and pharmacy systems. Despite hefty HL7 standards [22], in practice many messages of the same type vary considerably owing to different interpretations of the data entry fields. Therefore, although HL7 messages are designed to impose structure on the communication of clinical data, fields designed for succinct or coded information often contain verbose and/or unnormalized data, while fields designed with more leniencies are often overused and contain data that could have been structured. Even if HL7 messages are used appropriately, much of the valuable data exists as unstructured text in segments for observations, instructions, and notes, or in other attached documents. Therefore, we plan to use TRACE to enrich various clinical repositories and Master Data Management (MDM) systems with data mined from HL7 messages. Moreover, state-of-the-art clinical MDM systems include patient similarity algorithms [23]. We plan to integrate TRACE into our own comprehensive patient similarity algorithm, which will aid the cohort identification process and thus improve both operational analytics and patient care. For institutions with a cumbersome volume of patient records, TRACE may be easily deployed on a Hadoop cluster since SystemT is fully integrated with IBM's BigInsights, and thus may automatically leverage map-reduce algorithms.

In addition, data mined from unstructured clinical text may be used to create a true 360˚ view of a patient that is best visualized with a dashboard. We have developed a patient-centric dashboard that aggregates data from various clinical repositories and online medical information sources. We will enhance this view by extracting clinical concepts and mapping them to medical concept codes, which will serve as integration points. Drug references and prescription attributes may be used to search for side effects and drug interactions, while test results may be highlighted as important findings and aggregated with a patient's personal and family history.

VIII. Conclusion

Diverse, unstructured clinical text is an abundant source of information that, when accurately mined, may improve the operation of healthcare services in terms of efficiency, cost, and patient care. A hierarchical rules-based extractor such as TRACE and a pipeline of annotators such as cTAKES are potential NLP solutions for this task. However, for the stunted, ungrammatical phrases that are common in prescription documents, we have found that a simple hierarchical approach is satisfactory. Furthermore, since TRACE implements extensive rules-based optimization and workload sharing, it is an attractive alternative to cTAKES for massive-scale clinical text mining.

References

[1] "Request for proposals for a enterprise document sharing (XDS) registry and audit trail and node authentication (ATNA) repository solution," Alberta Health Services, Edmonton, 2011.

[2] G. K. Savova, et al., "Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES): architecture, component evaluation and applications," J. Am. Med. Inform. Assoc., vol. 17, no. 5, pp. 507–513, 2010.

[3] D. Baldwin and D. Carrell, "Rapidly deployable, highly scalable natural language processing using cloud computing and an open source NLP pipeline," Open Health NLP Whitepaper, Group Health Research Institute, Seattle, WA, 2010. [Online]. Available: https://wiki.nci.nih.gov/download/attachments/63996717/Rapid_Scalable_NLP_And_Cloud_Computing.doc?version=1&modificationDate=1320409953000.

[4] R. Krishnamurthy, et al., "SystemT: a system for declarative information extraction," SIGMOD Record, vol. 37, no. 4, pp. 7–13, 2008.

[5] D. Burdick, et al., "Extracting, linking and integrating data from public sources: a financial case study," IEEE Data Eng. Bull., vol. 34, no. 3, pp. 60–67, 2011.

[6] B. Alexe, et al., "Surfacing time-critical insights from social media," Proc. ACM SIGMOD International Conference on Management of Data, ACM, 2012, pp. 657–660.

[7] D. Cameron, V. Bhagwan and A. P. Sheth, "Towards comprehensive longitudinal healthcare data capture," 2012 IEEE International Conference on Bioinformatics and Biomedicine Workshops (BIBMW), IEEE, 2012, pp. 240–247.

[8] "HL7 RDE Message–Pharmacy/Treatment Encoded Order Message," Corepoint Health, 2014. [Online]. Available: http://www.corepointhealth.com/resource-center/hl7-resources/hl7-rde-message-pharmacy.

[9] L. Chiticariu, et al., "SystemT: an algebraic approach to declarative information extraction," Proc. 48th Annual Meeting of the Association for Computational Linguistics, ACL, 2010, pp. 128–137.

[10] "openNLP," Apache Software Foundation, 2010. [Online]. Available: http://opennlp.apache.org/.

[11] J. C. Reynar and A. Ratnaparkhi, "A maximum entropy approach to identifying sentence boundaries," Proc. Fifth Conference on Applied Natural Language Processing, ACL, 1997, pp. 16–19.

[12] A. Ratnaparkhi, "Learning to parse natural language with maximum entropy models," Mach. Learn., vol. 34, no. 1–3, pp. 151–175, 1999.

[13] G. Divita, A. C. Browne and T. C. Rindflesch, "Evaluating lexical variant generation to improve information retrieval," Proc. AMIA Symp., AMIA, 1998, pp. 775.

[14] B. L. Humphreys and D. A. Lindberg, "The UMLS project: making the conceptual connection between users and the information they need," Bull. Med. Libr. Assoc., vol. 81, no. 2, p. 170, 1993.

[15] W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper and B. G. Buchanan, "A simple algorithm for identifying negated findings and diseases in discharge summaries," J. Biomed. Inform., vol. 34, no. 5, pp. 301–310, 2001.

[16] G. Savova, et al., "Automated discovery of drug treatment patterns for endocrine therapy of breast cancer within an electronic medical record," J. Am. Med. Inform. Assoc., vol. 19, no. e1, 2012, pp. e83–e89, doi: 10.1136/amiajnl-2011-000295.

[17] S. J. Nelson, K. Zeng, J. Kilbourne, T. Powell and R. Moore, "Normalized names for clinical drugs: RxNorm at 6 years," J. Am. Med. Inform. Assoc., vol. 18, no. 4, pp. 441–448, 2011.

[18] K. W. Fung, C. McDonald and B. E. Bray, "RxTerms–a drug interface terminology derived from RxNorm," AMIA Annu. Symp. Proc., AMIA, 2008, pp. 227.

[19] "OmniPage Capture SDK," Nuance, 2014. [Online]. Available: http://www.nuance.com/for-business/by-product/omnipage/csdk/index.htm.

[20] G. Hripcsak, A. S. Rothschild, "Agreement, the f-measure, and reliability in information retrieval," J. Am. Med. Inform. Assoc., vol. 12, no. 3, pp. 296–298, 2005.

[21] S. M. Meystre, "Extracting information from textual documents in the electronic health record: a review of recent research," Yearb. Med. Inform., vol. 35, pp. 128–44, 2008.

[22] "Introduction to HL7 Standards," Health Level Seven International, 2014. [Online]. Available: http://www.hl7.org/implement/standards/index.cfm?ref=nav.

[23] "Enabling operational healthcare analytics with IBM InfoSphere Master Data Management," IBM Software Information Management Whitepaper, IBM Corporation, 2012. [Online]. Available: http://public.dhe.ibm.com/common/ssi/ecm/en/imw14615usen/IMW14615USEN.PDF.